

ASHITHA VENU MUDILIKULAM

High Resolution Image Processing Expert

as3707mu-s@student.lu.se | +32 498 656243 | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

PROFESSIONAL SUMMARY

Geospatial analyst with an Erasmus Mundus-funded MSc in Geo-information Science and Earth Observation for Environmental Modelling and Management, specialized in high-resolution image processing for agri-food sector. Previous hands-on experience at VITO evaluating Gross Primary Productivity models against AmeriFlux in-situ data, complemented by very high-resolution multisource field margin mapping in Belgium, and SAR-based mangrove biomass estimation. Proficient in Python, GIS (ArcGIS Pro, QGIS), statistical analysis, and satellite/UAV image processing (SNAP, Agisoft Metashape Professional), with a peer-reviewed publication in progress.

AREAS OF EXPERTISE

High-Resolution Image Processing • Remote Sensing • Data Science • Big Data Analysis • Python • Statistical Analysis • Machine Learning • Random Forest Classification • UAV/Drone Imagery • Satellite Image Processing • Google Earth Engine • SNAP • ArcGIS Pro • QGIS • Agisoft Metashape Professional • Raster & Vector Analysis • Time-Series Analysis (TIMESAT) • GPS Data Collection • OpenStreetMap (OSM) • LiDAR • Community Surveys

RELEVANT PROJECTS

Field Margin Quantification and Characterization Using Very High-Resolution Multisource Geospatial Data (Master's thesis)

- Quantified and characterized agricultural field margins in Wallonia, Belgium, by processing UAV imagery, orthophotos, LiDAR, land cover, and vector datasets at very high resolution.
- Applied spatial metrics to quantify margin extent, connectivity, and fragmentation across 30 sites, validating results against UAV data.

Burned Area Mapping in Northern Portugal

- Developed and validated a burned area mapping approach using Sentinel-2 imagery and Random Forest classification (full details in portfolio).

Machine Learning-Based Air Quality Prediction

- Estimated surface air pollutants using machine learning with Sentinel-5P, MODIS AOD, ERA5 meteorology, and OpenAQ data, producing national-scale pollution maps for data-scarce regions (full details in portfolio).

WORK EXPERIENCE

GIS Intern

July – August 2025

VITO, Mol, Belgium

Project: Evaluation of GPP Model with AmeriFlux In-Situ GPP Data

- Analyzed and evaluated Gross Primary Productivity (GPP) models by integrating remote sensing data with in-situ measurements from the AmeriFlux network.

Research Assistant

Feb 2024 – Jul 2024

College of Forestry, Kerala Agricultural University (KAU)

Project: Remote sensing based analysis of Urban Heat Island (UHI) and mapping hotspots of major Municipal Corporations in Kerala

- Prepared Land Surface Temperature (LST) and Land Use Land Cover (LULC) maps for 6 municipal corporations using ArcGIS.

Research Intern

Dec 2022 – Dec 2023

Indian Institute of Space Science and Technology (IIST)

Project: Biomass estimation of Mumbai mangroves using optical and microwave satellite remote sensing

- Utilized SNAP, ArcGIS, QGIS, and PolSARpro for radar interaction analysis and biomass data extraction from satellite imagery.

Intern

Aug 2022 – Sep 2022

Kerala Forest Research Institute (KFRI)

Project: Landslide Hazard Mapping of districts in Kerala

- Used Sentinel-2A and ASTER satellite data for preparation of thematic layers in ArcGIS for hazard mapping.

Project Assistant

Nov 2021 – Dec 2021

Kites Foundation – Haritha Keralam Mission

Project: Perunthode-Valiyathode Eco-restoration Project

- Conducted a scientific study involving watershed mapping and community opinion surveys for eco-restoration planning.

EDUCATION

MSc in Geo-information Science and Earth Observation for Environmental Modelling and Management

Lund University, Sweden & UCLouvain University, Belgium

- Funded by the prestigious Erasmus Mundus Scholarship

B.Sc. – M.Sc. (Integrated) Climate Change Adaptation

College of Climate Change and Environmental Science (CCCES), Kerala Agricultural University (KAU), India

RESEARCH AND PUBLICATIONS

Journal Article (In Progress)

Mudilikulam, A. V., Lakshmanan G. (2026). Above ground biomass assessment of mangroves using optical and microwave satellite remote sensing. *Journal of Coastal Conservation*.

Conference Presentation

Mudilikulam, A. V. (2023). Landslide hazard mapping of districts in Kerala. *Science Undergraduate Research Conference (SURC)*, 8–9 December 2023, Azim Premji University, Bengaluru.

CERTIFICATIONS

- Intermediate Python – DataCamp, February 2026
- Training on FME – GIM, October 2025
- Python for Data Science, AI & Development – IBM, July 2025
- UAVs in Precision Agriculture – University of Twente (ITC), February 2025
- GIS for Climate Action – ESRI Academy, December 2023
- Certificate in Environmental Studies (CES) – IGNOU, June 2022

SCHOLARSHIPS AND AWARDS

- **Erasmus Mundus Scholarship** – Awarded for MSc in Geo-information Science and Earth Observation for Environmental Modelling and Management

EXTRACURRICULAR INVOLVEMENT

University Union Councillor

2019 – 2020

Kerala Agricultural University

- Served as student representative at university level, advocating for college issues and student interests.

Fridays for Future – Volunteer

2018

- Organized and led climate strikes in Thrissur, promoting climate action and environmental awareness.